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Online Class Incremental Learning of Acoustic Events Using Acoustic Pre-trained Models

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***Abstract*—Performance of machine learning models trained in static settings degrades when they are confronted with unseen classes. In real-word scenarios, data is most often non-stationary and unavailable at once, prompting the development of Machine Learning (ML) algorithms - mostly DNN-based - that can learn incrementally and adapt to new data as it becomes available. However, many still suffer from catastrophic forgetting while learning new classes. To mitigate this, various Class-Incremental Learning (CIL) techniques have emerged, particularly in computer vision. Recently, Pre-Trained models (PTMs), known for their transfer learning capabilities, are increasingly used in CIL. In the acoustic domain, vision-based PTMs are commonly adapted, while acoustic PTMs remain underused despite their strong performance in acoustic classification tasks. Additionally, online versions of many of the traditional ML algorithms have been developed, yet their role in CIL is less explored. This study evaluates seven acoustic PTMs with four online ML algorithms on ESC-10, ESC-50, and UrbanSound8K datasets. Results show that, with strong PTMs, even simple classifiers like online Naive Bayes can outperform state-of-the-art methods in acoustic CIL.**

***Index Terms*—Online machine learning, continual learning, incremental learning, examplar-free CIL, examplar-based CIL, audio classification.**

I. Introduction

Online class-incremental learning (OCIL) enables machine learning models to learn sequentially from non-stationary data streams, in contrast to traditional offline batch learning. In many real-world applications, data is not available all at once, and retraining from scratch is often computationally infeasible. Classical class-incremental learning (CIL) involves learning from mini-batches of disjoint classes across tasks, aiming to acquire new knowledge while retaining previous knowledge.

OCIL introduces three core challenges: learning from one sample at a time, avoiding catastrophic forgetting [1], [2], and minimizing intransigence [3]. Broadly, CIL techniques fall into three main categories: regularization-based methods [4], experience replay [5], and parameter-isolation approaches [6]. While most CIL techniques originated from computer vision [7] field, they are increasingly being adapted to the acoustic domain [8].

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Recently, interest has grown in using large pre-trained deep learning models—especially DNN-based architectures as deep audio feature extractors (DAFEs) for OCIL. However, most studies rely on image-based PTMs whereas acoustic-specific PTMs remain underexplored [7]. Moreover, online variants of traditional machine learning algorithms have emerged as effective classifiers for streaming data and can complement the PTMs, yet their role in CIL remains largely overlooked.

This work aims to address this gap by evaluating seven acoustic PTMs as DAFEs in an OCIL setting using online machine learning classifiers. Our contributions are:

1) Comparison of four online ML algorithms using seven acoustic PTMs as DAFEs on three benchmark datasets: ESC-10, ESC-50 [9], and US8K [10].
2) Empirical evidence that acoustic PTMs offer robust and generalizable features for OCIL/CIL, similar to their vision-based counterparts.
3) Demonstration that online traditional ML models can effectively learn from a data stream and remain viable for OCIL tasks.
4) Extensive experimentation on ESC-10, ESC-50, and US8K show significant improvements over state-of-the-art methods, achieving average accuracies (AA) of 99.10%, 93.04%, and 80.69%, and percentage decrease in forgetting rate of 88.78%, 71.75%, and 9.53%, respectively.
5) Demonstration that with strong acoustic PTMs, even simple online ML algorithms like online Naive Bayes can outperform existing CIL approaches.

To the best of our knowledge, this is the first study to comprehensively evaluate both acoustic PTMs and online ML algorithms in the context of class-incremental learning.

II. Related Works

Several empirical studies in computer vision [11] and Natural Language Processing [12] suggested and encouraged the use of PTMs in CIL as it benefits downstream continual learning due to the strong knowledge transfer and resilience to catastrophic forgetting that the PTMs provide [7]. Zhou et al. [13] argued that generalizability provided by PTMs and adaptivity provided by adapted models are all that is needed for a successful CIL. They acknowledged the predominance of the visual recognition PTM-based CIL and the burgeoning interest in expanding beyond visual recognition.

Expressing such interest in the acoustic domain, a handful of studies leveraged PTMs for CIL. Sun et al. [14] proposed a Feature Generative Replay model for Environmental Sounds (FGR-ES) using ResNet18 and Wasserstein GAN (WGAN), achieving 64.91% and 78.40% accuracy on ESC-50 and US8K respectively. Bayram and Ince [15] integrated novelty detection with class-incremental learning using various PTMs (e.g., ResNet18, VGG16), anomaly detectors (e.g., KNN, GMM, OCSVM, AE), and CIL SOTA (e.g., LwF, iCarl, FearNet), achieving strong results with VGG16, GMM, and FearNet on ESC-10, ESC-50, and US8K datasets. For multi-label audio tasks, Mulimani and Mesaros [16] employed PANN-CNN14 with cosine similarity-based distillation loss, preserving performance across incremental phases. Pian et al. [17] proposed AV-CIL using AudioMAE and VideoMAE, showing gains on a novel audio-visual dataset.

These promising results of the acoustic PTMs emphasizes the importance of exploring others, particularity those that exhibited compelling performance in Acoustic Event Recognition (AER) and Acoustic Scene Classification (ASC) tasks such as Kumar [18], OpenL3 [19], PANNs [20], Vggish [21], and YAMNet [22].

This review highlights that visual PTMs and neural networks tend to dominate acoustic OCIL research, leaving many of the SOTA acoustic PTMs underutilized. Our study evaluates seven acoustic PTMs as feature extractors combined with four online ML algorithms, demonstrating that these combinations are effective and viable for acoustic class-incremental learning.

III. METHODOLOGY

A. OCIL Problem Definition

The objective of OCIL in our study is to train a model in a supervised manner to learn from a stream of dataset D one instance at a time through a sequence of tasks T, where each task consists of a set of classes that do not overlap with classes in other tasks. Let T be the total number of tasks in the OCIL procedure. The dataset $\mathcal{D}$ is partitioned into training and test subsets for each task $t \in \{1, 2, \ldots, T\}$, denoted by $\mathcal{D}_t^{\text{Tr}}$ and $\mathcal{D}_t^{\text{Ts}}$, respectively.

For each task t, we define:

$$\mathcal{D}_t^{\text{Tr}} = \{(x_i, y_i)\}_{i=1}^{N_t}, \quad \mathcal{D}_t^{\text{Ts}} = \{(x_i, y_i)\}_{i=1}^{M_t} \tag{1}$$

where:

- N_t and M_t denote the number of training and test samples in task t,
- (x_i, y_i) is a training or test instance with input $x_i \in \mathbb{R}^d$ and label $y_i \in \mathcal{Y}_t$,
- $\mathcal{Y}_t$ is the label space specific to task t.

In the disjoint CIL setting, it is assumed that the class labels across tasks do not overlap:

$$\mathcal{Y}_t \cap \mathcal{Y}_{t+1} = \emptyset, \quad \forall t \in \{1, 2, \ldots, T-1\} \tag{2}$$

Training Protocol

During training on task t, only the dataset $\mathcal{D}_t^{\text{Tr}}$ is accessible:

- In an exemplar-free CIL, no data from previous tasks is stored or revisited.
- In an exemplar-based CIL, a small subset of previous task samples (exemplars) is stored and made available in future tasks.

Evaluation Protocol

After each task t, the model is evaluated on all seen classes from tasks 1 through t. The cumulative label space is:

$$\mathcal{Y}_{1:t} = \bigcup_{j=1}^{t} \mathcal{Y}_j \tag{3}$$

Accordingly, the model is evaluated on the combined test sets:

$$\bigcup_{j=1}^{t} \mathcal{D}_j^{\text{Ts}}$$

B. Deep Audio Features Extractors

We evaluate seven pre-trained deep audio feature extractors (DAFEs), all trained on AudioSet [1], a large-scale dataset with approximately 2 million YouTube clips spanning 527 sound classes. All models use log mel spectrograms as input, except PANN Wavegram-Logmel, which combines wavegrams and spectrograms. Default configurations are used, except for OpenL3, where we set a hop size of 0.5 and an embedding size of 512.

Kumar et al. (KM) [18] introduced a 12-layer CNN accepting various sampling rates and producing 1024-dim embeddings.

OpenL3 [19] includes two variants—Music (OM) and Environment (OE)—generating (N, 512) dim embeddings where N is the number of frames in the audio clip.

PANNs [20] include CNN14-16K (PC) and Wavegram-Logmel (PW), both with a 14-layer CNN, outputting 2048-dim embeddings, and accepting audio sampled at 16khz, and 32khz respectively.

VGGish (VG) [21] uses a 5-layer VGG-based CNN to generate (N, 128) embeddings where N equals to the number of 0.96-second frames in the audio clip.

YAMNet (YM) [22] uses a MobileNet-based architecture with 25 depth-wise separable convolutional layers, producing (N, 1024) embeddings where N equals the number of 0.48-second frames in the audio clip.

C. Online Learning Classifiers

Online ML algorithms process data incrementally, making them fast and memory-efficient. After evaluating several from the River library [23], we selected four that performed well for acoustic event classification: Bernoulli Naive Bayes (BNB), Gaussian Naive Bayes (GNB), K-Nearest Neighbors (KNN), and Hoeffding Tree (HFT).

1) Online Bernouilli Naive Bayes: Online BNB [23] is an incremental version of the traditional Bernoulli Naive Bayes (BNB) classifier, designed to handle binary feature vectors in streaming or batch-incremental learning scenarios. Bernoulli Naive Bayes (BNB) assumes binary features and computes the probability that a given feature is present or absent in a class. At its core, Naive Bayes assumes that all features are independent of each other. Online BNB updates the model incrementally with each new training instance or mini-batch and stores the number of samples seen for each class and the number of times a feature was present in the samples from each class.

2) Online Gaussian Naive Bayes: Online GNB [23] is the incremental version of the standard Gaussian Naive Bayes (GNB) classifier. It assumes that each feature follows a Gaussian (normal) distribution and maintains the number of samples seen for class, the running mean for each feature, and the running variance for each feature.

3) Online K-Nearest Neighbors: Online K-NNN [23] is a version of the regular K-NN technique that was developed for contexts where data is streamed and learned over time. In contrast to storing the entire training dataset, online K-NN maintains a fixed or dynamic buffer where few training samples are stored. A first-in first-out (FIFO) strategy is most often used to update the instances in the buffer. As in the regular K-NN, it performs distance-based inference over the stored samples. The algorithm classifies a new instance by finding its k nearest neighbors and taking a majority vote.

4) Hoeffding Tree: The HFT Tree Classifier [23] is an incremental decision tree algorithm designed for real-time data stream mining. It builds decision trees using a small amount of memory and time per sample and makes statistically sound decisions about splitting nodes based on the Hoeffding inequality to determine, with high probability, when sufficient data has been seen at a node to make an optimal splitting decision. This enables the algorithm to learn from data in a single pass without storing all examples.

D. Methodology Summary

We denote the model $\phi(\cdot)$ as a combination of a feature extractor $F(\cdot)$ and a classifier $C(\cdot)$ and expressed as $\phi(\cdot)$ = $C(F(\cdot))$. We follow a transfer learning setup where the output of $F(\cdot)$ is taken from its fully connected layer without finetuning and represents a feature vector of dimension Eb = (f,n) where f is the number of frames in the audio clip and n the number of neurons in the fully connected layer. When necessary, the dimension of the feature vector is reduced through average pooling to a size of (1,n).

In our setup, we freeze $F(\cdot)$ and only update $C(\cdot)$ during the online training through the T defined tasks using the extracted features. During the evaluation, the predicted label $\hat{y}_t$ of an audio clip X_t at time t is denoted as:

$$\hat{y}_t = C(F(X_t)) \tag{4}$$

IV. Results and Discussion

A. Datasets

We conducted our experiments on three benchmark datasets for acoustic event classification: ESC-50, ESC-10 [9], US8K [10], all containing a mix of indoor and outdoor environmental sounds.

ESC-50 consists of 2,000 audio clips of 5 seconds each, sampled at 44.1 kHz, evenly distributed across 50 classes and organized into 5 predefined folds.

ESC-10 is a 10-class subset of ESC-50 with 400 clips comprising of sounds like baby cry, chainsaw, clock tick, dog bark, fire crackling, helicopter, person sneeze, rain, rooster, and sea waves.

US8K contains 8,732 clips across 10 classes, ranging from 0.3 to 5 seconds in length, sampled at 44.1, 48, or 96 kHz, and organized into 10 folds.

Unlike prior CIL studies that randomly split folds, we use strict K-fold cross-validation where training and test folds do not overlap. Following common CIL practices, we adopt the Train-From-Scratch (TFS) strategy [24], where classes are evenly divided across tasks: for a dataset with C classes and T tasks, each task contains C/T classes. For each task, the data set is randomly shuffled and split into training and test sets according to the folds.

Before embedding extraction (128 dimensions), audio clips were downsampled to 16 kHz for YAMNet, VGGish, OpenL3, Kumar, and PANN CNN14-16K, and to 32 kHz for PANN Wavegram-LogMel. Clips were zero-padded if shorter than 1 second.

B. Experiment Setup

We implemented the models using Keras [25], PyTorch [26], and the River [23] library for online machine learning. The experiments were run on a Razor X Lambda Tensorbook equipped with Intel 12th Gen i7 CPUs and 65GB RAM. We conducted extensive evaluations across 28 models, 3 datasets, 10 train-test splits, and 10 repetitions per setup, totaling 8,400 runs.

C. Evaluation Metrics

We utilized four of the commonly used metrics in CIL to assess the performance of our models. The Top-1 accuracy at the end of each incremental task Ti, indicated as A_t, the accuracy in the last incremental task, denoted as A_L and the average accuracy, denoted as AA. A_L gives a measure of the overall accuracy of the model among all the classes whereas AA provides a measure of the overall accuracy across all the incremental tasks and is calculated as .

$$AA = \frac{1}{T}\sum_{t=1}^{T} A_t \tag{5}$$

Where:

- AA is the task-averaged value of the accuracy A_t.
- T is the total number of tasks.
- A_t is the value of the accuracy at task t.

Also, we employed the final performance dropping rate PD = A_1 - A_L which is the difference between the accuracy in the first task A_1 and the accuracy in the last task A_L. PD tries to quantify how much forgetting happens in the overall procedure and a smaller PD is preferable.

D. Comparison to previous research

We evaluated our models on the datasets comparing their performance to existing class-incremental learning (CIL) methods using the accuracies and the performance drop metrics.

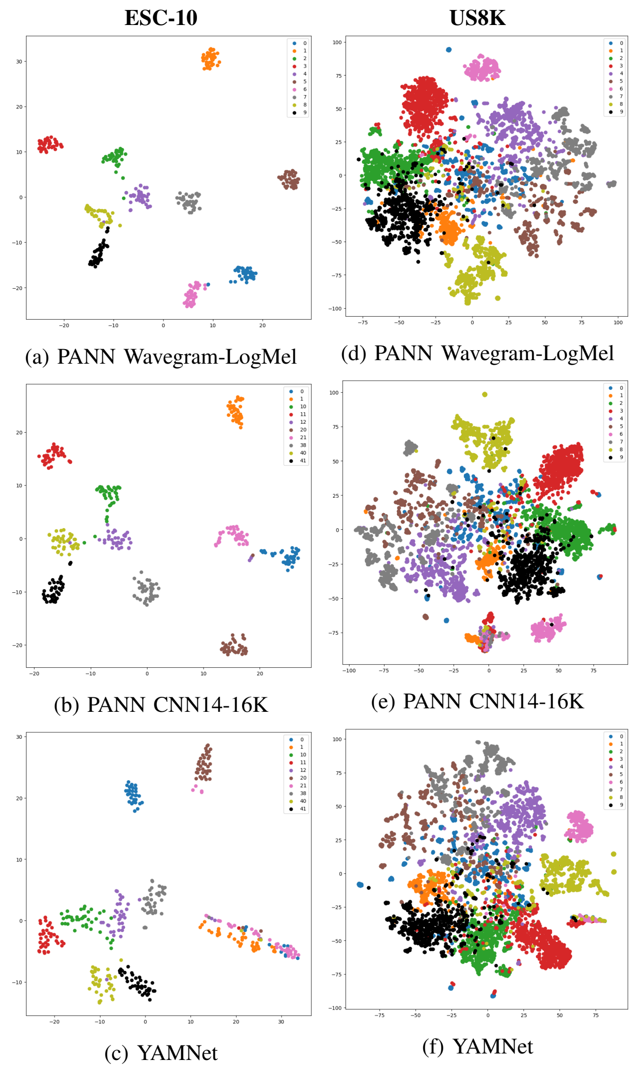
1) ESC-10 Results: As shown in Table I, our models significantly outperform prior methods. Our best model, combining PANN Wavegram-Logmel with KNN (PW-KNN), achieves 99.10% average accuracy and a 1.59% performance drop—representing improvement rates of 18.90% in AA and of 88.78% in PD over the state-of-the-art (SOTA) methods. Among the classifiers, KNN performed best, closely followed by Bernoulli Naive Bayes, while PANN Wavegram-Logmel provided the most robust audio features among the PTMs (Fig. 1a). The small performance variation across tasks as observed in Table I and Fig. 2a demonstrates the model's ability to retain prior knowledge and incorporate new classes without catastrophic forgetting.

TABLE I: Comparison with prior works across each incremental task on ESC-10 dataset. We format the <u>**first**</u>, **second** and <u>third</u> best results.

Methods	Accuracy in each task (%) 1	2	3	4	5	AA↑	PD↓
Lwf [5]	89.58	21.43	18.75	11.11	10.00	30.17	79.58
Gem [27]	89.58	64.29	45.31	26.39	38.75	52.86	50.83
Ewc [4]	89.58	51.79	25.00	19.44	23.75	41.91	65.83
Wa [28]	89.58	71.43	54.69	58.33	68.75	68.56	20.83
Foster [6]	89.58	76.79	71.88	75.00	72.50	77.15	17.08
FGR-ES [29]	91.67	86.46	79.86	81.25	77.50	83.35	14.17
PW-BNB	99.59	98.78	98.50	98.04	<u>97.81</u>	<u>98.55</u>	<u>1.78</u>
PC-KNN	99.69	99.02	98.64	98.27	**98.13**	**98.75**	<u>**1.56**</u>
PW-KNN	99.91	99.63	99.05	98.60	<u>**98.31**</u>	<u>**99.10**</u>	**1.59**

2) ESC-50 Results: ESC-50 results show similar trends. Our best-performing model, PW-GNB, achieves 93.04% average accuracy and a 7.45% performance drop—surpassing the existing best model with improvement rates of 43.34% in AA and of 71.75% in PD. Unlike with ESC-10, Naive Bayes classifiers outperformed KNN in this experiment, highlighting their robustness when paired with PANN-based embeddings (Table II).

3) US8K Results: On the US8K dataset, our models still outperform existing approaches, though with more modest gains. PW-GNB achieves 80.69% average accuracy and 18.79% PD as shown in Table III, representing improvement rates of 2.92% and 9.53% over the SOTA. As depicted in Fig.4, the variations in AA across all incremental tasks are more pronounced although slightly better than those of the SOTA as displayed in Table III. This suggests that the PTMs struggled to generate well-separated embeddings for US8K classes (Fig. 1d, 1e, and 1f), making their classification more challenging.

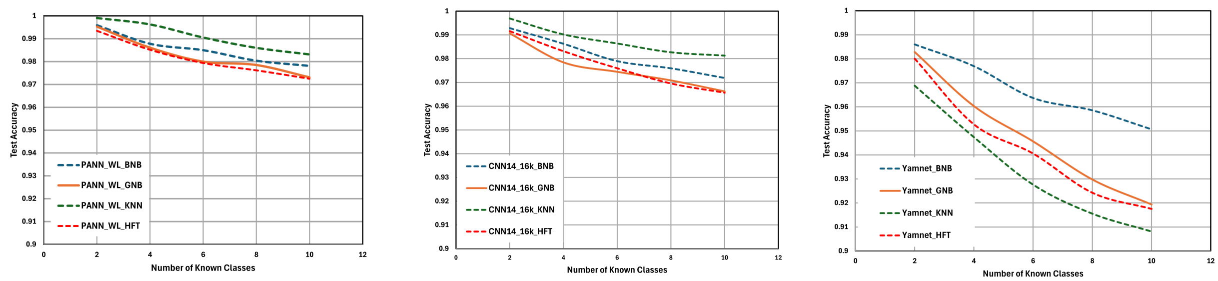


(a) PANN Wavegram-LogMel (d) PANN Wavegram-LogMel

(b) PANN CNN14-16K (e) PANN CNN14-16K

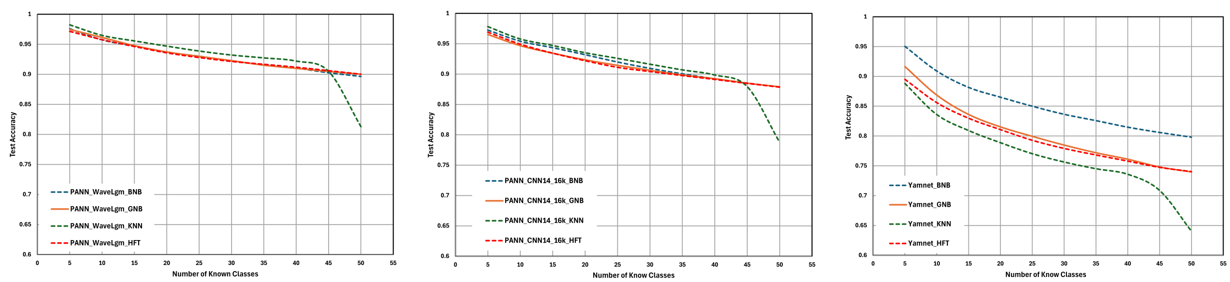
(c) YAMNet (f) YAMNet

Fig. 1: Class Embedding Space representation for ESC-10 and US8K with the top-3 PTMs.



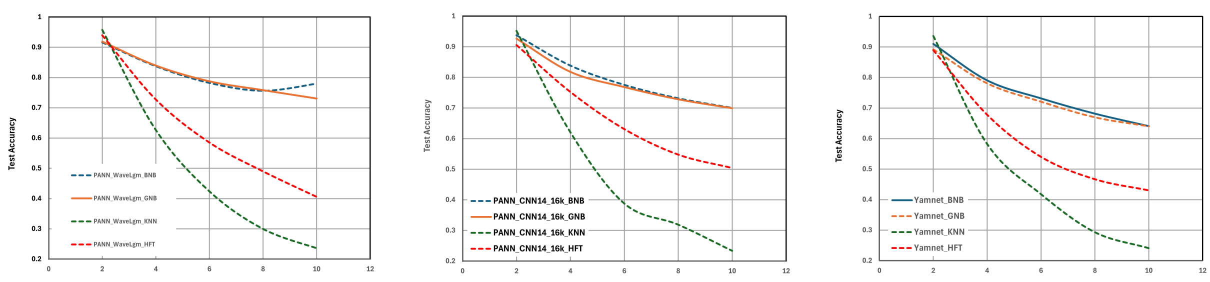
(a) PANN Wavegram-LogMel (b) PANN CNN14-16K (c) YAMNet

Fig. 2: Test accuracy results comparison with the top-3 DAEEs and the online algorithms for ESC-10 dataset.



(a) PANN Wavegram-LogMel (b) PANN CNN14-16K (c) YAMNet

Fig. 3: Test accuracy results comparison with the top-3 DAFEs and the online algorithms for ESC-50 dataset.



(a) PANN Wavegram-LogMel (b) PANN CNN14-16K (c) YAMNet

Fig. 4: Test accuracy results comparison with the top-3 DAFEs and the online algorithms for US8K dataset.

TABLE II: Comparison with prior works across each incremental task on ESC-50 dataset.We format the <u>**first**</u>, **second** and <u>third</u> best results.

	Accuracy in each task (%)												
Methods	1	2	3	4	5	6	7	8	9	10	11	AA↑	PD↓
Lwf [5]	80.82	54.30	35.29	23.61	19.41	17.50	11.61	12.50	10.87	10.94	7.25	25.82	73.57
Gem [27]	80.42	24.61	16.18	9.72	6.58	10.31	7.14	9.66	8.70	4.17	2.50	16.36	77.92
Ewc [4]	80.42	25.00	16.18	22.92	11.84	9.69	13.10	15.06	9.24	4.17	3.75	19.22	76.67
Wa [28]	80.83	75.78	67.65	66.10	60.68	62.94	60.83	55.62	56.68	54.84	52.00	63.09	28.83
Foster [6]	81.25	75.00	69.49	67.71	62.50	60.62	61.31	57.95	58.15	55.73	50.50	63.66	30.75
FRG-ES [29]	85.42	76.67	72.64	61.88	61.92	60.07	60.71	55.78	58.98	60.71	59.05	64.91	26.37
PW-GNB	97.46	96.14	94.77	93.73	92.98	92.28	91.52	90.98	90.48	90.01	-	<u>**93.04**</u>	**7.45**
PW-BNB	97.57	95.73	94.70	93.67	93.02	92.23	91.58	91.04	90.27	89.66	-	**92.95**	<u>7.91</u>
PW-HFT	97.15	95.77	94.63	93.58	92.79	92.15	91.66	91.18	90.60	90.01	-	**92.95**	<u>**7.14**</u>

Notes: Our study used 10 incremental tasks instead of the 11 in the previous work.

TABLE III: Comparison with prior works across each incremental task on US8K dataset.We format the **<u>first</u>**, **second** and <u>third</u> best results.

Methods	Accuracy in each task (%) 1	2	3	4	5	AA↑	PD↓
Lwf [5]	89.48	68.84	58.77	51.33	47.94	63.27	41.54
Gem [27]	89.48	66.64	63.01	54.76	52.46	65.27	37.02
Ewc [4]	89.29	67.43	60.34	51.75	49.08	63.58	40.21
Wa [28]	89.76	70.25	63.77	58.19	54.47	67.29	35.29
Foster [6]	89.61	83.33	76.86	70.76	63.54	76.82	26.07
FRG-ES [29]	90.52	83.56	75.36	72.83	69.75	78.40	<u>20.77</u>
PW-GNB	91.86	83.94	78.74	75.83	**73.06**	**<u>80.69</u>**	**18.79**
PW-BNB	91.60	83.72	78.19	75.60	**<u>73.30</u>**	**80.48**	**<u>18.30</u>**
PC-BNB	93.78	83.86	77.55	73.15	<u>70.02</u>	<u>79.67</u>	23.77

V. Conclusion

This paper presented online class-incremental learning (CIL) algorithms that integrate large acoustic pre-trained models (PTMs) as deep audio feature extractors with online machine learning classifiers. We evaluated seven acoustic PTMs and four online classifiers across 28 model variants on three benchmark datasets. Results indicate that PANN and YAMNet produced the most robust embeddings, with Naive Bayes classifiers delivering top performance across most classification tasks. These findings position acoustic PTMs as strong alternatives to vision-based models in acoustic CIL. Notably, several of our methods surpass state-of-the-art baselines, highlighting their effectiveness and resilience to catastrophic forgetting. Future work will investigate the plasticity-stability trade-off in Naive Bayes classifiers and explore the use of ensembles of PTMs.

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